Stochastic Sampled-Data Security Synchronization Control for Delayed Reaction-Diffusion Neural Networks under Multi-Source Deception Attacks

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Abstract—This paper focuses on the stochastic sampled-data (SD) security synchronization control for delayed reaction-diffusion neural networks (RDNNs) subject to multi-source deception attacks under spatially point measurements (SPMs). The SD security synchronization scheme, which takes into account the communication limitations of multi-source deception attacks and stochastic sampling, is based on SPMs and only available at a finite number of fixed spatial points. Based on the Lyapunov functional (LF) and inequality techniques, the mean square exponential stability of the synchronization error system (SES) is ensured by some synchronization criteria, which are established via a stochastic SD security controller under SPMs and expressed as linear matrix inequalities (LMIs). Lastly, two examples are provided to demonstrate the effectiveness of the presented stochastic SD security synchronization control scheme.